Survey on Value at Risk in Financial Crisis in 2008

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1. Introduction

In the aftermath of the financial crisis in 2008, public began to examine the various factors leading to the calamity. Among some critical factors, such as subprime mortgages, credit ratings, and etc., the risk management of the financial institutes obviously had played crucial role during the crisis. To be specific, the most widely used risk management tool --- Value at Risk (VaR) should be held responsible for the crisis to some extent.

VaR is a standard financial tool used to measure the market risk of financial portfolios. Pioneered in 1993, VaR is a part of the "Weatherstone 4:15 pm" daily risk assessment report, in the RiskMetrics model at J. P. Morgan.

Later by 1995, the Basel Committee on Banking Supervision proposed amendment to the 1988 Basle Capital Accord "The Supervisory Treatment of Market Risks". It proposed that financial institutes should use either an internal methodology or the Bank for International Settlements (BIS) standard methodology to computer VaR and meet capital requirements based on VaR estimates.

Until BIS issued the Market Risk Amendment to the Basel II Capital Accord of 2004, VaR was required to be calculated daily, using a "99th percentile, one-tailed confidence interval". Banks were also allowed to choose their own models, but the recent financial crisis scenario makes us wonder what happened to the VaR and which models should we choose in future.

2. Assumptions of the conventional VaR models

2.1 VaR Methodology

Duffie and Pan (1997) introduces some basic ideas about VaR. VaR is a tool to analyze risk, while risk can be broadly defined as the degree of uncertainty to the future net

return. There are mainly four risks that financial institutions have to deal with: 1) credit risk, which is due to the counterparty's failure to meet the obligation; 2) operation risk, which caused by payment or transaction faults; 3) liquidity risk, which refers that entities sell low liquidity assets with large discount to deal with the large negative cash flow occurred in a short time; 4) market risk, which refers to the uncertainty of future profits caused by the changes of market conditions. Among them, market risk is of great importance, since it reflects the potential economic loss of the market prices decline of asset portfolios. The risk that VaR analyzes is market risk.

According to the course of Risk Management by Prof. Neil Pearson in Business School, VaR is defined as the amount that an organization can expect to lose less than x percent of the time (if positions are left unchanged for t days). This measure emphasizes three variables: the potential loss of a portfolio, the probability of the loss, and the time in which this loss occurs. Take a look at an example borrowed from Prof. Pearson. "Probability of 5 percent, horizon of 1 day, VaR of $1 million" means that the organization expects to suffer daily losses of $1 million or more on only 5 percent of business days.

For this definition, there are several things we need to look at. First, why do we measure the risk of a PORTFOLIO? It is because financial institutions are facing hundreds and thousands of risk from assets and market factors, there are too many deltas to estimate and monitor. Second, it is a probability measure, meaning that we place an upper bound on losses in the sense that they will exceed the VaR with only a small target probability which is usually between 1% and 5%. Third, it measures risk of losses over short durations, assuming a NORMAL market of volatility.

VaR's problem, perhaps rooted deeply within the basic structure of financial economics is the two important assumptions: 1) the market is efficient; and 2) the risk of financial market is normally distributed.

To further illustrate this, we need to scrutinize some conventional VaR models' assumptions.

Although different models have different calculation methods, the basic steps are almost the same: 1) estimate the probability distribution of portfolio return; 2) calculate VaR or other inferences. Based on the differences on step 1, the existing models are mainly divided into three categories: parametric model, nonparametric model, and semiparametric (or semi-nonparametric) model. However, the classification is actually not very strict. For example, RiskMetrics is commonly treated as parametric model, but some also place it in the class of nonparametric. Moreover, some treat CAViaR as semiparametric model, while others call it nonparametric. We don't discuss it here, but use parametric and nonparametric classification.

2.2 Parametric Model

This methodology assumes certain parametric specification for price behaviors. The early VaR estimation was based on the return distribution of parametric model, such as normal distribution or t-distribution. Later, more sophisticated models based on ARCH or GARCH were invented, under the trademarks of RiskMetrics, Creditmetrics, and KMV, which can fit some characteristics of financial data, such as fat-tail and data dependency.

2.2.1 GARCH(1,1)

For GARCH (1,1), there are two main issues: 1) the specification of variance, 2) the assumption that standardized residuals are independent, identical distributed (iid). The first one is due to the characteristic of financial data, that is, the financial return series subject to data dependence. The second one is for the convenience of parameter estimation. To further calculate GARCH, we need to choose one distribution for the residuals. Usually, normal distribution is chosen. And then we estimate the parameters by using likelihood functions and calculate 5% quantile using time series of estimated variances.

2.2.2 RiskMetrics

For RiskMetrics, its variance is calculated by Exponentially Weighted Moving Average, which corresponding to Integrated GARCH Model and the decay factor (lamda) is usually set as 0.94 or 0.97, as suggested by J.P.Morgan for computing daily volatilities and for calculating monthly volatilities (defined as a holding period of 25 days). Also, normality assumption is made for the standardized residuals.

2.2.3 Pros and cons of parametric model

The advantages of parametric model are 1) the result of parameters can be easily interpreted; 2) parametric techniques have higher statistical efficiency for estimating quantiles, when parametric models fit well with the return data.

However, parametric model is subject to the specification mistakes. This method tends to underestimate VaR, since the normality assumption of residuals is not consistent with financial price behaviors. The specification of variance equation is also under question.

Thanks to the abundant researches for the financial price behaviors, most economists agree on the following characteristics of financial data:

1) Volatility clustering, i.e. the high autocorrelation of absolute and squared returns;

2) Large kurtosis, meaning the density of unconditional return distribution has fatter tails and has higher peak than the normal distribution;

3) Mild skewness of the returns.

Providing reasonable assumptions of financial data is very crucial to the methodology.

2.3 Nonparametric Model

2.3.1 Historical Simulation

The most common VaR estimation is Historical Simulation. Proposed by Dowd (2001), it is based on sample quantile and enormously simplified the calculation, because it doesn't assume return distribution but relies on rolling windows. There are many variations of this methodology. The most popular one is the naïve Historical Simulation (HS), and the most successful one is the Filtered Historical Simulation (FHS).

For HS, first we choose an observation window (or sample size), from 6 months to 2 years, and then ascendingly order the return (or profit/loss) within the window. With this series, we can get θ quantile, on the left of which there are θ% of observations and on the right there are 1- (θ%) of observations. To calculate the second day's VaR, we move the window by one observation and then repeat the process.

2.3.2 Filtered Historical Simulation

For FHS, location-scale model is used to prefilter the data. It is a generalized historical simulation. It possesses all the good characteristics and immunes of many weaknesses of the HS. In FHS, the assumption of stationary is relaxed, and historical returns are standardized by volatility estimated on the particular day, producing approximately iid residual. VaR forecasts are then generated by computing the VaR from paths simulated using draws from the filtered residuals. Barone-Adesi, Giannopoulos, and Vosper (1999, 2002) and Pritsker (2001) show that this method performs rather well.

2.3.3 Hybrid Model

Hybrid Model is a variation of HS proposed by Boudoukh, Richardson and Whitelaw (1998) and combines both RiskMetrics and HS by multiplying exponential declining

weights. Compared with HS, Hybrid Model has specification with more flexibility and dynamics, which relieving the stress for choosing the window (or sample size) and being able to response quickly to the large losses. However, there is no guidance for choosing the weight. Moreover, rapidly declining weights is similar to the situation of a small sample, while slowly declining weights will use data from a distant past.

2.3.4 Pros and cons of nonparametric model

Nonparametric model is better than parametric one due to the following reasons. First, it is easy to implement. Second, it uses the empirical distribution of past returns to generate realistic future scenarios. Non-linear positions can be re-priced under each scenario. Also, historical simulation does not require the computation of a covariance matrix.

Yet, there are some problems with this methodology. First, although nonparametric model doesn't specify distribution for return, an underlying assumption is that the return distribution within the sample remains unchanged. If all the returns in the sample have the same distribution, then all the returns in this time series have this exact distribution.

Second, choosing sample size needs tradeoff. It is obvious that only a large sample size can guarantee the significance of the statistical estimation. However, this will increase modeling biases and then the weights for the most recent observations will be too small, leading to a smooth VaR curve. As a result, in nonparametric model, a compromise is required for smoothing parameter selection (Fan and Gijbels, 1996).

Third, if the market is moving from a relative low volatile state towards a relative high volatile state, then VaR will be underestimated, otherwise, it will be overestimated. Furthermore, the extreme dispersed return values will lead to jumps. Specifically, a huge negative value will push VaR to jump upward, but after a time of sample size, VaR will jump downward accordingly.

2.4 Still use VaR?

VaR is still appealing not only to financial industry but also to supervisors, notwithstanding. For one thing, it is simple enough – only a single number and no more. What makes it more charming is that it can be applied to almost all the asset class and can measure both individual and firm-wide risk.

So, in a short time period, VaR won't disappear, but will probably be in another form. In recent years, computing power has increased substantially. Thus, estimating more sophisticated models induces only little higher computation costs. Maybe we can expect them to be common VaR estimation approaches.

3. Literature Review

During and after the crisis, professionals and scholars widely discussed about why didn't we forecast the crisis and how can we improve. However, the specific papers discussing the improvement of VaR to better deal with period of high volatility are handful. One of the most interesting suggestions came from McAleer et al. (2012), which compares the performance of several VaR models during different stages of the crisis and then makes some combinations of different models to form aggressive and conservative risk management strategies.

The same idea is mentioned in Chen et al. (2012) at the end of their paper, which compares a range of well-known fully parametric models for pre, during, and post crisis time period, but to find that no model did constantly well across the different forecast horizons.

Therefore, our idea is that, since we couldn't find an ideal model which can outperform throughout all the stages of crisis and adapt to every level of volatility, then we should form a combination of promising models.

3.1 Combination of VaR models

Chen et al. (2012) compare a bunch of parametric models and find out that there is no single model can consistently outperform others across all the period of the crisis. Prior to the financial crisis, for 1-day-ahead forecasting, the GJRGARCH with skewed Student-t errors ranked best, followed by other asymmetric volatility models. During and after the crisis, skewness and fat tails dominate at the 1% level, with non-stationary models doing best at 5%. For 10-day-ahead forecasting prior to the crisis, the EGARCH models had the best performance. Generally, GARCH models dominated the SV models in forecast performance. They find that all the 10-day forecasting models underestimated risk levels during the crisis and were rejected for risk coverage during and after the crisis. Therefore, they suggested to employing combinations of prominent models as risk management strategy.

McAleer et al. (2012) proposed a new approach to model selection for predicting VaR, consisting of combining alternative risk models. They compared conservative and aggressive strategies for choosing between VaR models and found that the forecasts of VaR could be improved using a combination of models rather than a single model. They used daily capital charges (DCC) penalty proposed by Basel Accord II as a criteria to measure the efficiency of those models. Different strategies obviously yield different effects for DCC. An aggressive risk management strategy yielded the lowest mean DCC, and had the highest frequency of minimizing DCC, but also tended to violate too often. A conservative risk management strategy would have far fewer violations, and a correspondingly higher mean DCC. The area between the bounds provided by the aggressive and conservative risk management strategies would need some more investigation.

Although combination of forecasting models is quite common in the time series literature, it is rarely used for forecasting VaR for risk management, according to Chiriac and Pohlmeier (2010). They used different benchmark criteria for model comparison of course.

Therefore, if someone decides to form a combination of forecasting models, two questions would come up, that is, what are promising models and how to choose from them?

3.2 Promising models

The answer to the questions could be found in the piles of literatures which compared an enormous amount of advanced models and suggested that their models are the best. For example, Beder (1995) applies eight common VaR methodologies2 to three hypothetical portfolios. Their results show that the differences among these methods can be very large, with VaR estimates varying by more than 14 times for the same portfolio. Bao, Lee, and Saltoglu (2003), who examine and compare the forecasting ability of 13 distributional assumptions crossed with nine volatility structures, and Hansen and Lunde (1994), who consider 330 such models. Brooks and Persand (2003) did the comparison of various univariate models and a MGARCH for VaR and volatility forecasting.

Yet, there are still too many models to be compared and investigated.

3.2.1 Extreme Value Theory (EVT)

Recently, models based on Extreme Value Distributions (EVDs) are very popular and are often used to deal with the fat-tail problem of financial data. EVD uses Generalized Pareto Distribution to fit the extreme tail part of the data. This EVT is based on Balkema-de Haan-Pickands Theorem. This method adapt to iid returns and dependent returns which can be expressed as iid innovations.

EVT seems to be a very general approach to tail estimation. The main advantage of it is that the use of a GEV distribution to parameterize the tail doesn't seem to be a very restrictive assumption, since it covers most of the widely used distributions. However,

there are also some problems to be considered. First, the assumption of iid observations seems to be not very consistent with the characteristics of financial data. Although generalizations to dependent observations have been proposed, they either estimate the marginal unconditional distribution or impose conditions that rule out the volatility clustering behavior typical of financial data. Second, EVT works only for very low probability levels. How low these probability levels must be is hard to tell on a priori ground. The results in Brooks et al. (2005) favor the EVT and GARCH approaches, though their conclusions are based only on the 5% VaR using a holdout period of only 250 days.

3.2.2 Conditional Autoregressive Value at Risk (CAViaR) + EVT

The Conditional Autoregressive Value at Risk, or CAViaR model was introduced by Engle and Manganelli (1999). Since then, it has been widely used for forecasting or benchmarking. The basic intuition is to model directly the evolution of the quantile over time, rather than the whole distribution of portfolio returns. The only assumption required under this framework is that the quantile process is correctly specified. In particular, no assumption on the distribution of the error terms is needed, hence reducing the risk of misspecification. Moreover, even if the quantile process is misspecified, the minimization of the regression quantile objective function can still be interpreted as the minimization of the Kullback-Leibler Information Criterion, which measures the discrepancy between the true model and the one under study.

The strategy of combining CAViaR and EVT is the following. First fit a CAViaR model to get an estimate of the quantile so that sensible estimates can be obtained. Then construct the series of standardized quantile residuals and impose an assumption. Under this assumption, EVT can be applied to the standardized quantile residuals to get an estimate of the tail. Engle and Manganelli (2001) proved that CAViaR + EVT outperforms under the situation of heavy-tailed data generation process.

3.2.3 Recent Innovations

In addition, more sophisticated VaR models are keeping emerging. A¨ıt-Sahalia and Lo (2000) introduced the concept of economic valuation of VaR and compared it with the statistical VaR. Cai and Wang (2008) suggest to estimate VaR and Expected Shortfall using a new nonparametric VaR estimator, combining theWeighted NadarayaWatson (WNW) estimator of Cai (2002) and the Double Kernel Local Linear (DKLL) estimator of Yu and Jones (1998). Chen and Tang (2005) investigate nonparametric VaR estimation, when no regressors are present. Taylor (2008) proposes to combine double kernel quantile regression with exponential smoothing of the dependent variable in the time domain. 1% and 99% VaRs are predicted from the model along with some benchmarks, but extreme quantiles are not considered.

Even though every model has its own innovations, we are just looking for promising ones which could perfectly fit the characteristics of financial returns and have the potential to accommodate different levels of volatilities.

4. Conclusion

The serious recession caused by the financial crisis in 2008 led the public to question the risk management strategy of those financial institutes. As one of the most widely used risk management tool, VaR was discussed in many articles and journals. This term paper surveys on the papers discussing the performance of VaR during the financial crisis.

First, I analyzed the underlying assumption faults of the conventional VaR models and concluded that the iid and normality assumptions are not consistent with the characteristics of financial returns, thus undermining the predictions they made. Then I briefly introduced the recent papers about VaR in the financial crisis of 2008. Their main finding is that there is no single model is optimal over time, especially for the financial

crisis. One reasonable proposal is that we can form a combination of VaR forecast models to accommodate the changing level of return volatilities. Though there are lots of promising models with sophisticated mechanism to fit the characteristics of financial data, some well-defined criteria for choosing models are needed, and so do the specific performance evaluation criteria.

As far as I am concerned, we need to reflect on the risk management of financial institutes, not only to find some strategies to deal with the situation of high volatility, but also to find some better strategies aiming at outperform across all the periods of crisis and turbulence. Further researches should be some empirical study on forming a combination of advanced models, not limited to parametric models and GARCH family, and more importantly, try to find a set of comprehensive and effective criteria for choosing and evaluating the model performance. We also should consider the different time horizons from pre, during, and post crisis and different portfolios from different markets in different countries. More works should be done.

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